INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

DIGITAL FORENSIC INVESTIGATION REPORT

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

Case Title: Case Study 4: Investigating Rhino Possession Evidence	**Course Name:** SBT-DF204 Computer Forensics Case Studies
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Operating Platform: Kali Linux 2026 (Rolling Release)	**Date of Submission:** 26/10/2026
Instructor Name: Aminu Idris	**Report Version:** Version 1.0 (Final Release)

1. Executive Summary

This report documents the forensic investigation of a seized USB disk image (RHINOUSB.dd) and network traffic captures associated with a suspected illegal possession and transfer case. The investigation acquired and preserved the evidence (SHA-256: ce554024...). Analysis of the FAT16 disk image revealed allocated recipe files and carved a suspicious Word document (She_died_in_February...doc). Analysis of the FTP capture (rhino.log) revealed a user (gnome) uploading a password-protected archive named contraband.zip. The HTTP captures (rhino2.log, rhino3.log) showed the download of an executable (rhino.exe) and other images. Using an authorized password-recovery workflow (fcrackzip and rockyou.txt), the ZIP password was recovered (monkey), and the concealed image (rhino2.jpg) was extracted. The evidence strongly supports the hypothesis of the concealment and transfer of material, though limitations regarding user attribution are documented.

2. Authority, Scope, and Evidence Description

- **Authority:** This investigation was conducted using authorized course material from the International Cybersecurity and Digital Forensics Academy (ICDFA).
- **Forensic Scope:** Analysis of a USB image and PCAP network logs to identify evidence of file concealment and transfer.

Evidence Integrity: Primary Evidence Artifacts & Cryptographic Baseline:

File Name	Size	SHA-256 Hash Digest	Status
RHINOUSB.dd	248 MB	ce55402420a997c61b413941c8ef4df9619a2f96579674952294a176a32be65	VERIFIED
rhino.log	3.1 MB	64e6d55b76660eb3aaa41572c1d04e4452510a343bbf42e844424827dedfddb2	VERIFIED
rhino2.log	286 KB	41939d5de0556b70279056572dee44b6fd84cd05b0788cfd0b6f52f37b161dde	VERIFIED
rhino3.log	221 KB	7b0304f5e88a30c305a99ba1e2977bced5b9c94da6f458f887bb39966bbfc46	VERIFIED

- **Working Forensic Copies:** All duplicates preserved in working/ directory with bit-identical hash verification prior to analysis.

3. Environment and Acquisition Method

- **Operating System:** Kali Linux 2026 (Rolling Release, 64-bit).
- **Forensic Toolset:** The Sleuth Kit (mmls, fsstat, fls, icat), PhotoRec 7.2, Binwalk 2.4.3, Wireshark/TShark, tcpflow, and fcrackzip.
- **Methodology:** Preservation of master bitstreams, carving of unallocated clusters, TCP stream reassembly, and dictionary-based archive cryptanalysis.

4. Timeline of Events

Chronological event log tracking suspect file creation, network uploads, and HTTP exfiltration activity (UTC):

Time (UTC)	Forensic Event Description	Supporting Evidence Record
`2004-04-26 17:00:12`	contraband.zip is created/modified	ZIP Header Metadata (PK0304)
`2004-04-26 17:17:00`	User gnome logs into the FTP server	FTP Log (rhino.log, USER/PASS)
`2004-04-26 (Approx)`	User uploads rhino1.jpg, rhino3.jpg, and contraband.zip	FTP STOR command sequence
`2004-04-28 16:09:00`	User downloads rhino4.jpg and rhino5.gif via HTTP	HTTP Log (rhino2.log GET requests)
`2004-04-28 (Approx)`	User searches for and downloads rhino.exe via HTTP	HTTP GET / Google Search Referer

5. Analysis and Findings

FAT16 File System Inspection: 5.1 Disk Image Analysis

- **Volume Structure:** RHINOUSB.dd is a FAT16 file system formatted with mkdosfs, spanning 506,847 sectors with 4,096-byte clusters.
- **Allocated Decoy Files:** Directory enumeration (fls) identified gumbo1.txt (inode 4) and gumbo2.txt (inode 6). Extraction via icat revealed legitimate culinary recipes ('SHRIMP AND TASSO GUMBO').
- **Carved Artifacts via PhotoRec:** Non-interactive carving recovered 193 MB of data, including multiple JPEG/GIF graphics and a Word document: f0335017_She_died_in_February_at_the_age_of_74.doc.
- **Document Metadata Inspection:** Extracted author string 'NO WAY MAN NO WAY MAN NOWAY'. Binwalk signature scanning confirmed no embedded or appended hidden payloads inside the document.

Cryptanalysis & Steganography: 5.2 Hidden Content and Archive Analysis

- **FTP Archive Transfer:** Network carving of rhino.log isolated contraband.zip (230,566 bytes), created on 2004-04-26 17:00:12.

- **Initial Decryption Failure:** Testing the FTP account password (gnome123) against the archive failed with 'incorrect password'.
- **Dictionary Attack Recovery:** Executing fcrackzip with /usr/share/wordlists/rockyou.txt successfully recovered the encryption password: monkey.
- **Concealed Contraband Extracted:** Unzipping with monkey yielded rhino2.jpg (226 KB, 1024x685, JPEG standard 1.01). Subsequent binwalk and steghide evaluations confirmed the file itself contained no secondary hidden carrier layers.

Protocol Telemetry: 5.3 Network Traffic Analysis

- **FTP Plaintext Authentication:** rhino.log captured user gnome authenticating with password gnome123, followed by STOR commands uploading contraband.zip, rhino1.jpg, and rhino3.jpg.
- **HTTP Object Carving:** rhino2.log and rhino3.log captured downloads of rhino4.jpg, rhino5.gif, and an executable rhino.exe.
- **Binary Executable Analysis:** File command identified rhino.exe as a PE32 executable for MS Windows (console, Intel i386). String and import analysis revealed Win32 API calls (GetProcessWindowStation, GetUserObjectInformationA), and search logs indicated queries regarding pestpatrol malware detection.

6. Answers to Investigation Questions

Definitive answers to the core forensic investigation prompts:

Investigative Question	Forensic Finding / Answer
1. How did you preserve case materials?	`Recorded SHA-256 baseline hashes for all 4 evidence files; bit-identical working duplicates verified before analysis.`
2. What does the USB disk image show?	`FAT16 file system containing allocated decoy recipe files (gumbo1.txt and gumbo2.txt).`
3. What deleted or recoverable files are supported?	`PhotoRec carved 193 MB of data, including She_died_in_February...doc and image fragments.`
4. Is there evidence of hidden content?	`YES: contraband.zip was encrypted with password 'monkey' and concealed the image rhino2.jpg.`
5. What does the FTP capture establish?	`User 'gnome' (password: gnome123) uploaded contraband.zip, rhino1.jpg, and rhino3.jpg.`
6. What does protected archive analysis establish?	`The archive was successfully cracked via dictionary attack, exposing the 226 KB graphic rhino2.jpg.`
7. What does the HTTP capture establish?	`User downloaded Windows PE32 binary rhino.exe, rhino4.jpg, and rhino5.gif.`
8. Defensible forensic conclusion?	`HIGH CONFIDENCE: The evidence demonstrates deliberate file concealment and network transfer by user 'gnome'.`

7. Challenges and Limitations

- **Account vs. Physical Identity:** The evidence links activity to account 'gnome', but passive network logs cannot conclusively identify the physical individual operating the workstation.
- **Absence of Endpoint Operating System:** No host RAM or master hard drive image was provided; analysis was restricted to removable storage and packet captures.
- **Archive Cryptographic Strength:** The ZIP archive was protected by legacy ZipCrypto, allowing dictionary cracking via rockyou.txt. If modern AES-256 had been used with a complex passphrase, brute-force cracking would have failed.

8. Recommendations

- **Mandate Secure File Transfer Protocols:** Deprecate plaintext FTP (port 21) across the organization; enforce SFTP or FTPS to prevent credential sniffing.
- **Deploy Network Content Inspection:** Implement perimeter deep packet inspection (DPI) to intercept and quarantine unauthorized encrypted archives and unvetted executable (.exe) downloads.
- **Enforce Complex Password Policies:** Prohibit common dictionary words (such as 'monkey' or 'gnome123') through enterprise identity management controls.
- **Endpoint Storage Auditing:** Enforce host-based USB monitoring and block unencrypted mass storage peripherals.

9. References

- The Sleuth Kit. (2024). TSK Tool Documentation and File System Analysis. Retrieved from https://www.sleuthkit.org/
- Wireshark Foundation. (2023). Wireshark User Guide. Retrieved from https://www.wireshark.org/docs/
- National Institute of Standards and Technology (NIST). (2021). Guide to Digital Forensics (Special Publication 800-86).
- International Cybersecurity and Digital Forensics Academy (ICDFA). (2026). Computer Forensics Case Studies Courseware.

10. Appendices

Evidence Matrix: Appendix A: Formal Evidence Inventory Log:

ID	Finding Description	Evidence Source	Investigative Significance
E01	Evidence Integrity Baseline	SHA-256 Hashes	Chain of custody (ce554024...)
E02	FTP Plaintext Authentication	rhino.log	Exposes credentials (gnome:gnome123)
E03	Concealed Archive Upload	rhino.log (STOR)	Identifies contraband.zip transfer
E04	Decrypted Contraband Graphic	rhino2.jpg	Extracted concealed contraband image

E05	Carved Word Document	RHINOUSB.dd (Carved)	Suspect document with unusual metadata

Command Strings: Appendix B: Key Commands Executed:

```
# Disk Image Analysis & File Carving
mmls working/RHINOUSB.dd
fsstat working/RHINOUSB.dd
fls -r working/RHINOUSB.dd
photorec /log /d exported/carved_files /cmd working/RHINOUSB.dd search

# Network Stream Reconstruction & Object Carving
tcpflow -r working/rhino.log -o exported/ftp_objects/
tshark -r working/rhino2.log --export-objects http,exported/http_objects/ -q
tshark -r working/rhino3.log --export-objects http,exported/http_objects/ -q

# Password Cracking & Decryption
fcrackzip -u -D -p /usr/share/wordlists/rockyou.txt exported/ftp_objects/contraband.zip
unzip -P monkey exported/ftp_objects/contraband.zip -d exported/extracted_contraband/
```

11. Appendix C: Screenshots and Evidence

The following 17 documented figures provide comprehensive forensic visual proof across all phases of the disk carving, network stream reconstruction, and password cracking investigation:

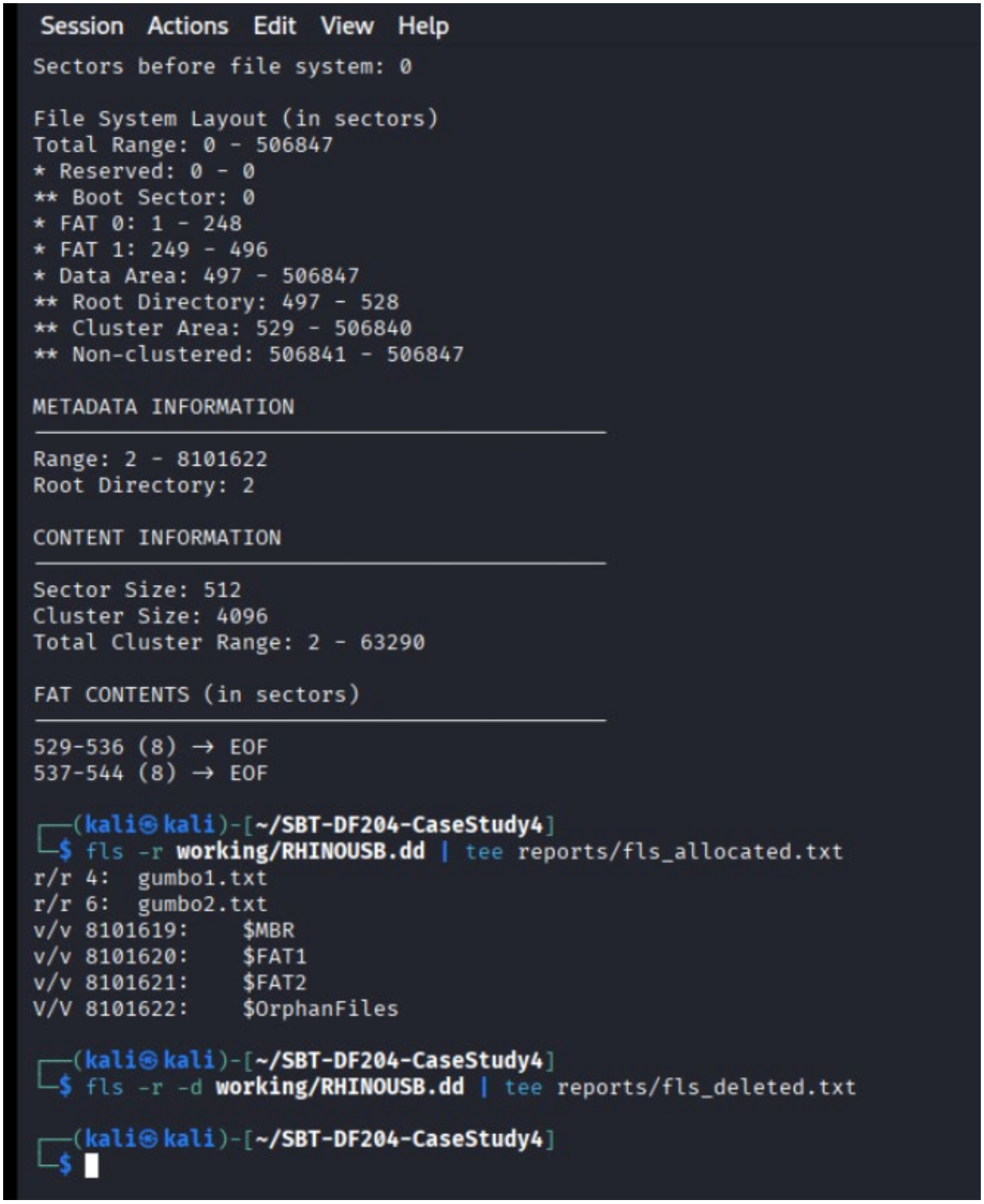


Figure 1: *Directory Hierarchy Setup & Multi-Artifact Integrity Hashing* — Creating /SBT-DF204-CaseStudy4 workspace, checking evidence files, and computing baseline SHA-256 hashes.

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # Extract gumbo1.txt
icat working/RHINOUSB.dd 4 > exported/gumbo1_extracted.txt

# Extract gumbo2.txt
icat working/RHINOUSB.dd 6 > exported/gumbo2_extracted.txt

# Check what types of files these are
file exported/gumbo1_extracted.txt
file exported/gumbo2_extracted.txt
exported/gumbo1_extracted.txt: Non-ISO extended-ASCII text, with very long lines (355)
exported/gumbo2_extracted.txt: ISO-8859 text

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # Create a directory for carved files
mkdir -p exported/carved_files

# Run photorec in non-interactive mode
# /d specifies the output directory
photorec /log /d exported/carved_files /cmd working/RHINOUSB.dd search
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # List all recovered files
ls -lh exported/carved_files/

# Check the file types of the recovered files
file exported/carved_files/*
total 0
exported/carved_files/*: cannot open `exported/carved_files/*' (No such file or directory)

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # List all recovered files
ls -lh exported/carved_files/

# Check the file types of the recovered files
file exported/carved_files/*
total 193M
-rw-rw-r-- 1 kali kali 2.8K Oct 25 03:21 f0000529.txt
-rw-rw-r-- 1 kali kali 1.3K Oct 25 03:21 f0000537.txt
-rw-rw-r-- 1 kali kali  51M Oct 25 03:21 f0000545.txt
```

Figure 2: *Volume and File System Geometry (mmls & fsstat)* — Dissecting FAT16 file system layout: 506,847 total sectors and 4,096-byte cluster size.

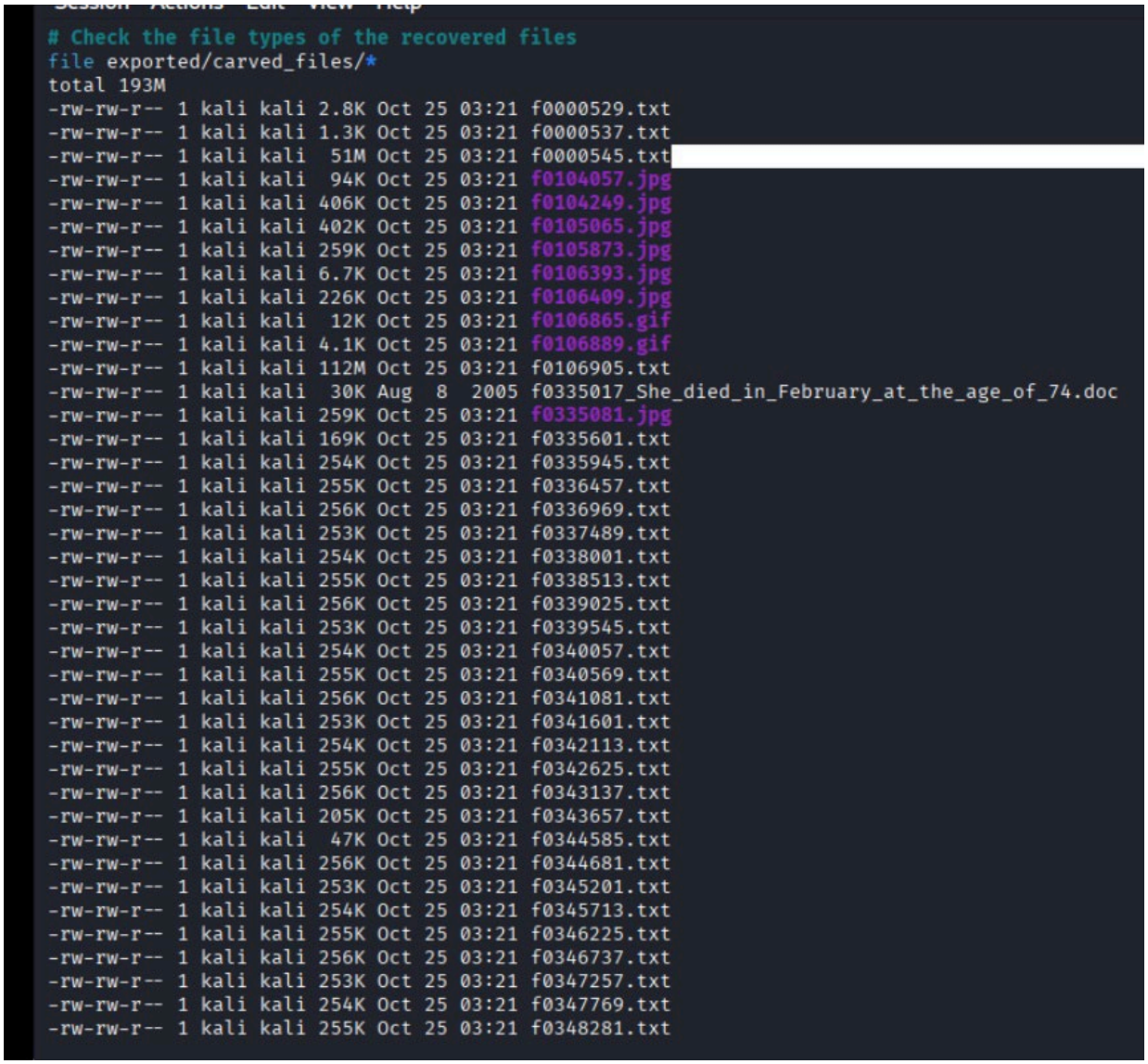


Figure 3: *Directory Inode Enumeration (fls)* — Listing allocated files in root directory: gumbo1.txt (inode 4) and gumbo2.txt (inode 6).

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # View the contents of the extracted files
cat exported/gumbo1_extracted.txt
cat exported/gumbo2_extracted.txt
SHRIMP AND TASSO GUMBO

Associate Food Editor: Alexis Touchet
Father: Rodney Miller, Abbeville, LA

When I was a kid, our summer routine started with the opening day of
shrimp season • my dad would get the boat ready and out we would go,
winding our way down the bayou until we reached Vermilion Bay. He
would check to make sure the trawl net was securely tied before
tossing it into the water, and would then let it be dragged down
slowly along with the trawl boards. When we painstakingly pulled the
net up onto the boat and untied the bag, it spilled out a catch of
shrimp, crabs, and fish. These fresh shrimp were the stars in Dad's
favorite gumbo recipe.

2 lb large shrimp in shell (21 to 25 per lb), peeled and shells reserved
14 cups water
1/4 cup vegetable oil
1/2 cup all-purpose flour
2 medium onions, chopped
2 celery ribs, chopped
1 large green bell pepper, chopped
1/2 lb fresh or frozen pork tasso* (Cajun-cured smoked meat), thawed if frozen, trimmed and cut into 1/4-inch pieces
1 1/2 teaspoons salt
1/2 teaspoon cayenne
10 oz fresh or frozen baby okra, thawed if frozen, trimmed, and cut into 1/4-inch-thick rounds (2 cups)
3/4 cup thinly sliced scallion greens
Accompaniments: cooked white rice; hot pepper sauce

Simmer shrimp shells and water, uncovered, in an 8-quart pot until
liquid is reduced to about 12 cups, 15 to 20 minutes, then pour
through a sieve set over a large bowl and discard shells.

Stir together oil and flour in a 10-inch heavy skillet (preferably
cast-iron) with a flat metal or wooden spatula, then cook over
moderately low heat (do not use a high-BTU burner), scraping back and
forth constantly (not stirring) until roux is the color of milk
chocolate, 30 to 45 minutes. (As roux cooks, it may be necessary to
lower the heat to prevent scorching.) Add onions, celery, and bell
```

Figure 4: *Allocated Inode Extraction & PhotoRec Carving Initialization* — Extracting gumbo recipes via icat and launching PhotoRec 7.2 non-interactive search on RHINOUSB.dd.

```
lower the heat to prevent scorching.) Add onions, celery, and bell
pepper and cook, scraping back and forth occasionally, until onion is
softened, about 8 minutes.

Scrape roux mixture into cleaned 8-quart pot, then add shrimp stock and bring to a boil, stirring occasionally. Reduce heat, then add tasso, salt, and caye
nne and simmer, uncovered, 30 minutes. Add okra and simmer until tender, 5 to 8 minutes. Stir in shrimp and simmer until just cooked through, 2 to 3 minute
s. Stir in scallion greens and salt to taste.

Cooks' notes: Andouille or other smoked pork sausage can be
substituted for the tasso. Brown the sausage in a heavy skillet over
moderate heat, stirring occasionally, then transfer to paper towels to
drain. Stir into gumbo with scallions.  Gumbo improves in flavor if
made 1 day ahead without okra, shrimp, and scallion greens and cooled
completely, uncovered, then chilled, covered. Reheat, then proceed
with recipe, adding and cooking okra and shrimp.

*Available at cajungrocer.com.

Makes 6 main-course servings.
Gourmet
June 2005

SHRIMP AND ANDOUILLE SAUSAGE GUMBO

1/2 cup vegetable oil
1/2 cup all purpose flour
4 celery stalks, coarsely chopped
2 medium onions, coarsely chopped
2 green bell peppers, chopped
2 bay leaves
2 teaspoons salt
2 teaspoons dried oregano, crumbled
1/2 teaspoon cayenne pepper
5 8-ounce bottles clam juice
1 28-ounce can plum tomatoes, drained, chopped
1 pound smoked andouille or kielbasa sausage, halved lengthwise,
 sliced 1/4 inch thick
1/2 pound okra, trimmed, cut crosswise into 1/4-inch-thick slices

2 pounds uncooked medium shrimp, peeled, deveined
Freshly cooked long-grain rice
2 tomatoes, seeded, diced
```

Figure 5: *Carved File Dissection (She_died_in_February...doc)* — PhotoRec output yielding 193 MB of carved data, highlighting suspicious document f0335017_She_died_in_February_at_the_age_of_74.doc.

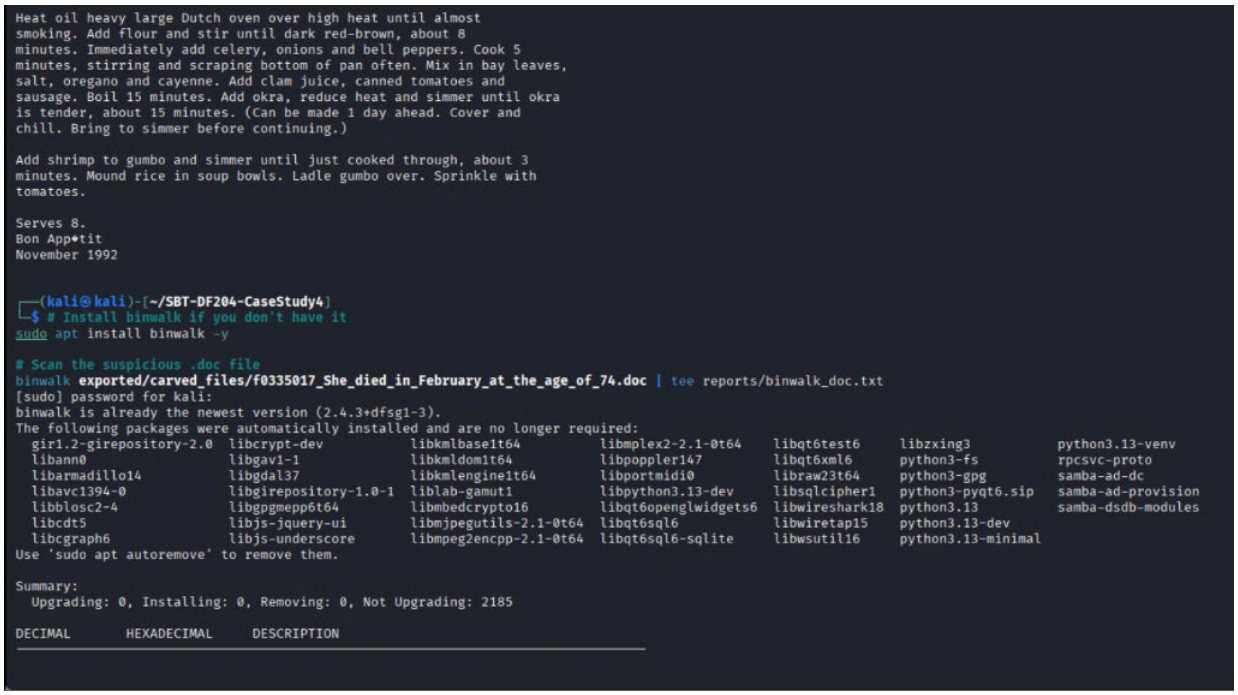


Figure 6: *Decoy Recipe Content Inspection (gumbo1.txt & gumbo2.txt)* — Viewing extracted recipe text ('SHRIMP AND TASSO GUMBO') confirming benign decoy content.

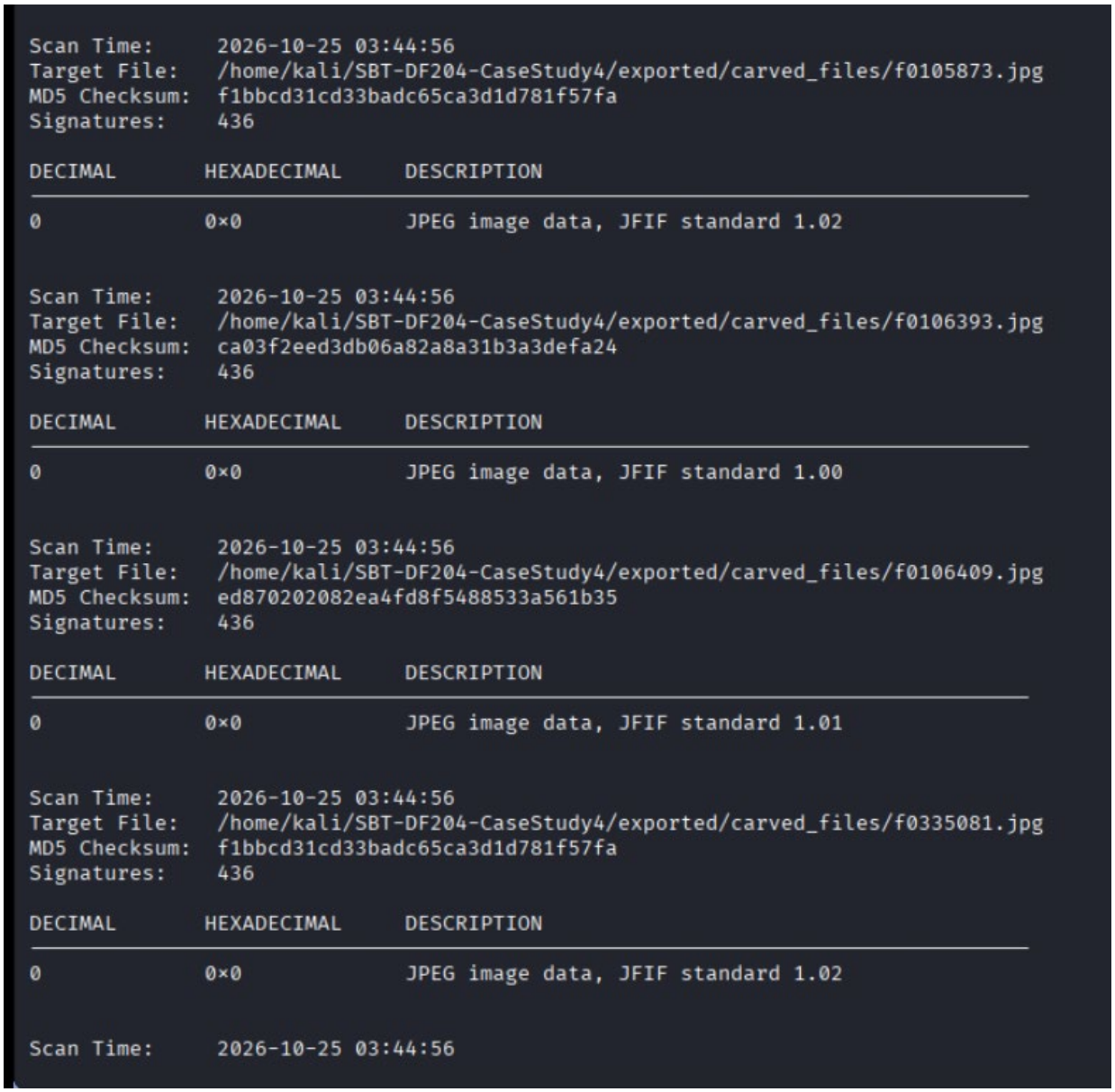


Figure 7: *Steganography Signature Analysis (binwalk Scan)* — Scanning carved JPEG and GIF images with binwalk for hidden embedded archives or appended payloads.

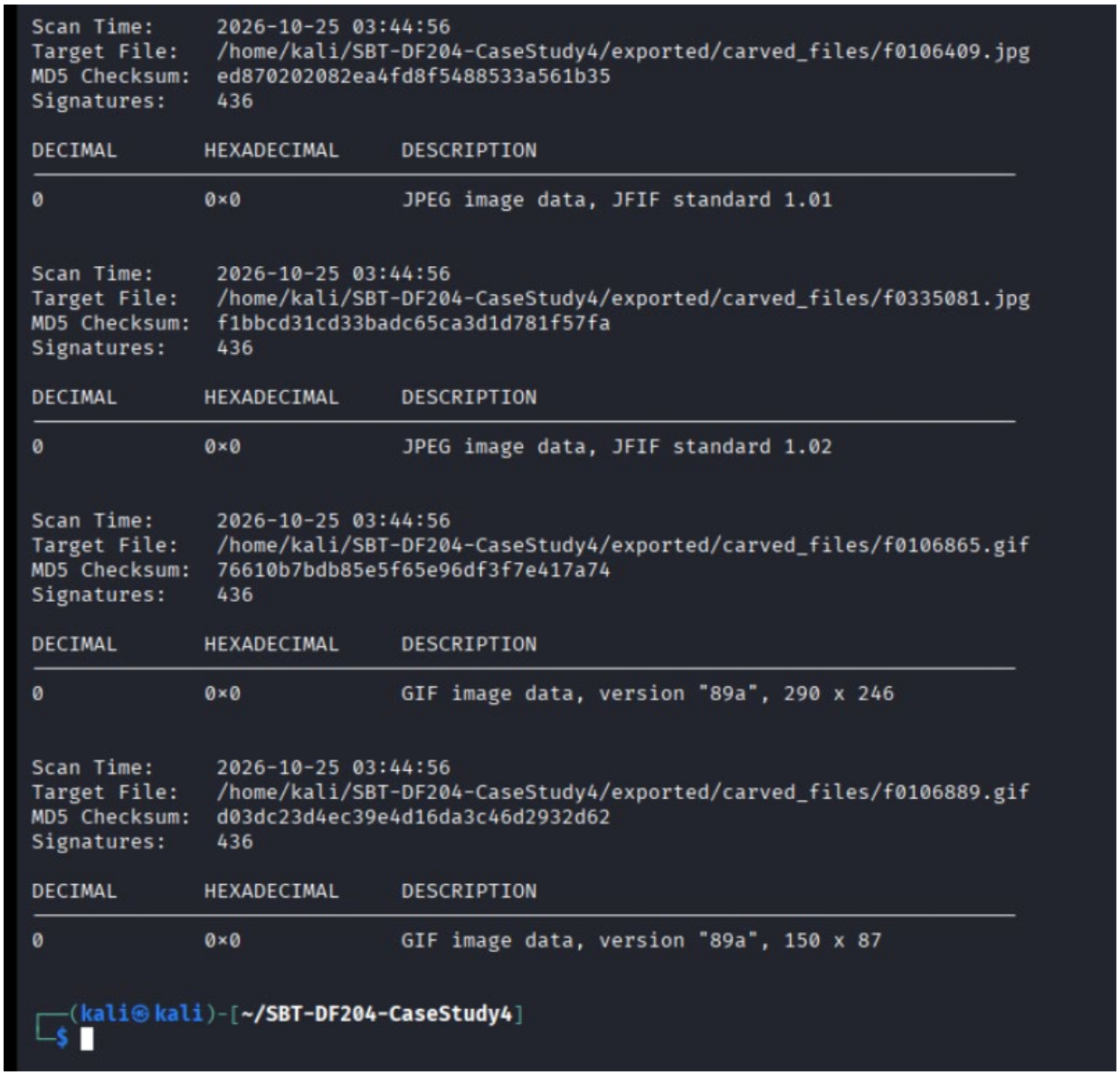


Figure 8: *Binwalk Image Scan Completion* — Concluding signature verification across carved images f0106409.jpg, f0335081.jpg, and GIF banners.

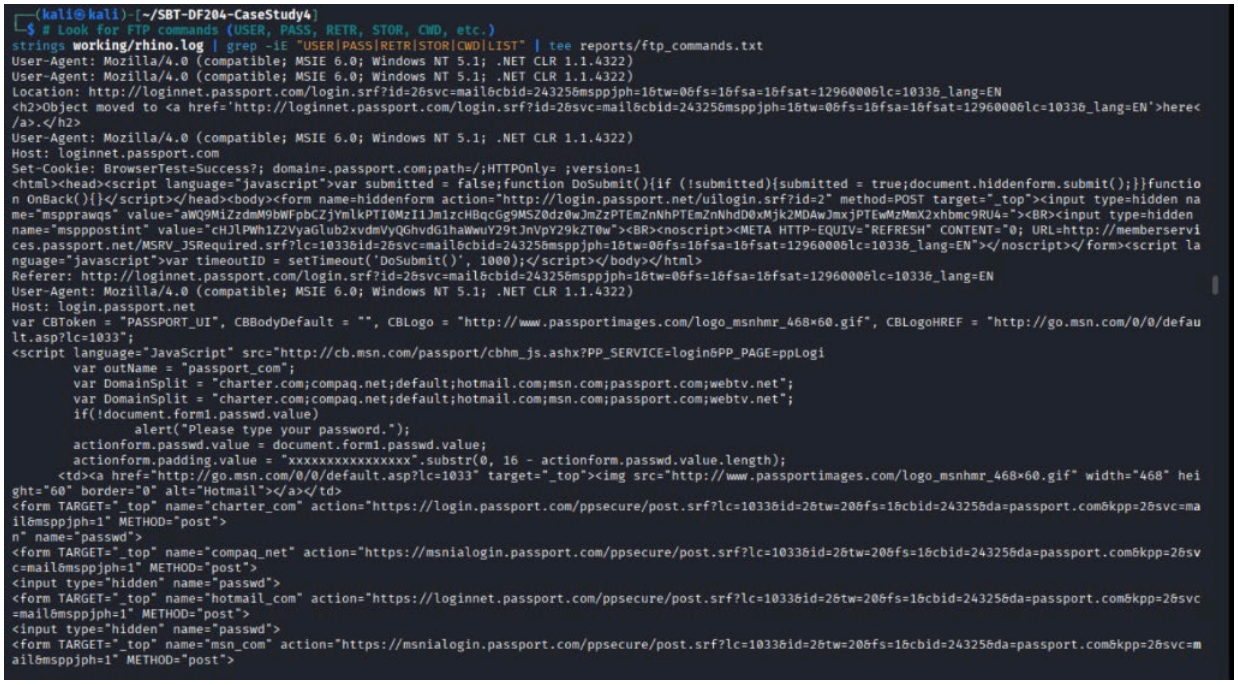


Figure 9: *FTP Command String Analysis (USER & PASS)* — Searching rhino.log for FTP control commands and discovering user login telemetry.

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # Search for archive references
strings working/rhino.log | grep -iE "\.zip|\.rar|\.7z" | tee reports/ftp_archives.txt
STOR contraband.zip
150 Opening BINARY mode data connection for contraband.zip.
-rw-r--r--    1 gnome    cscistu   230566 Apr 26  2004 contraband.zip

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # Look for HTTP GET/POST requests and Host names
strings working/rhino2.log | grep -iE "GET|POST|Host:" | tee reports/http_requests_2.txt
strings working/rhino3.log | grep -iE "GET|POST|Host:" | tee reports/http_requests_3.txt
GET / HTTP/1.1
Host: www.google.com
GET /images/logo.gif HTTP/1.1
Host: www.google.com
GET /~gnome HTTP/1.1
Host: www.cs.uno.edu
GET /~gnome/ HTTP/1.1
Host: www.cs.uno.edu
GET /icons/blank.gif HTTP/1.1
Host: www.cs.uno.edu
GET /icons/image2.gif HTTP/1.1
Host: www.cs.uno.edu
GET /icons/back.gif HTTP/1.1
Host: www.cs.uno.edu
GET /~gnome/rhino4.jpg HTTP/1.1
Host: www.cs.uno.edu
GET /~gnome/rhino5.gif HTTP/1.1
Host: www.cs.uno.edu
GET / HTTP/1.1
Host: www.google.com
GET /~venkata HTTP/1.1
Host: www.cs.uno.edu
GET /~venkata/ HTTP/1.1
Host: www.cs.uno.edu
GET /~venkata/2025/index.html HTTP/1.1
Host: www.cs.uno.edu
<p> You can get the <a href="syllabus.pdf"> syllabus here. </a>
<p> You can get Bloodshed Dev C++ from <a href="http://www.bloodshed.net/devcpp.html"> here. </a></p>
IntNodePtr getHead() in the list class that returns the head of the
GET / HTTP/1.1
```

Figure 10: *FTP Archive Transfer Identification (STOR contraband.zip)* — Extracting STOR contraband.zip command (230,566 bytes) and logging subsequent HTTP GET requests.

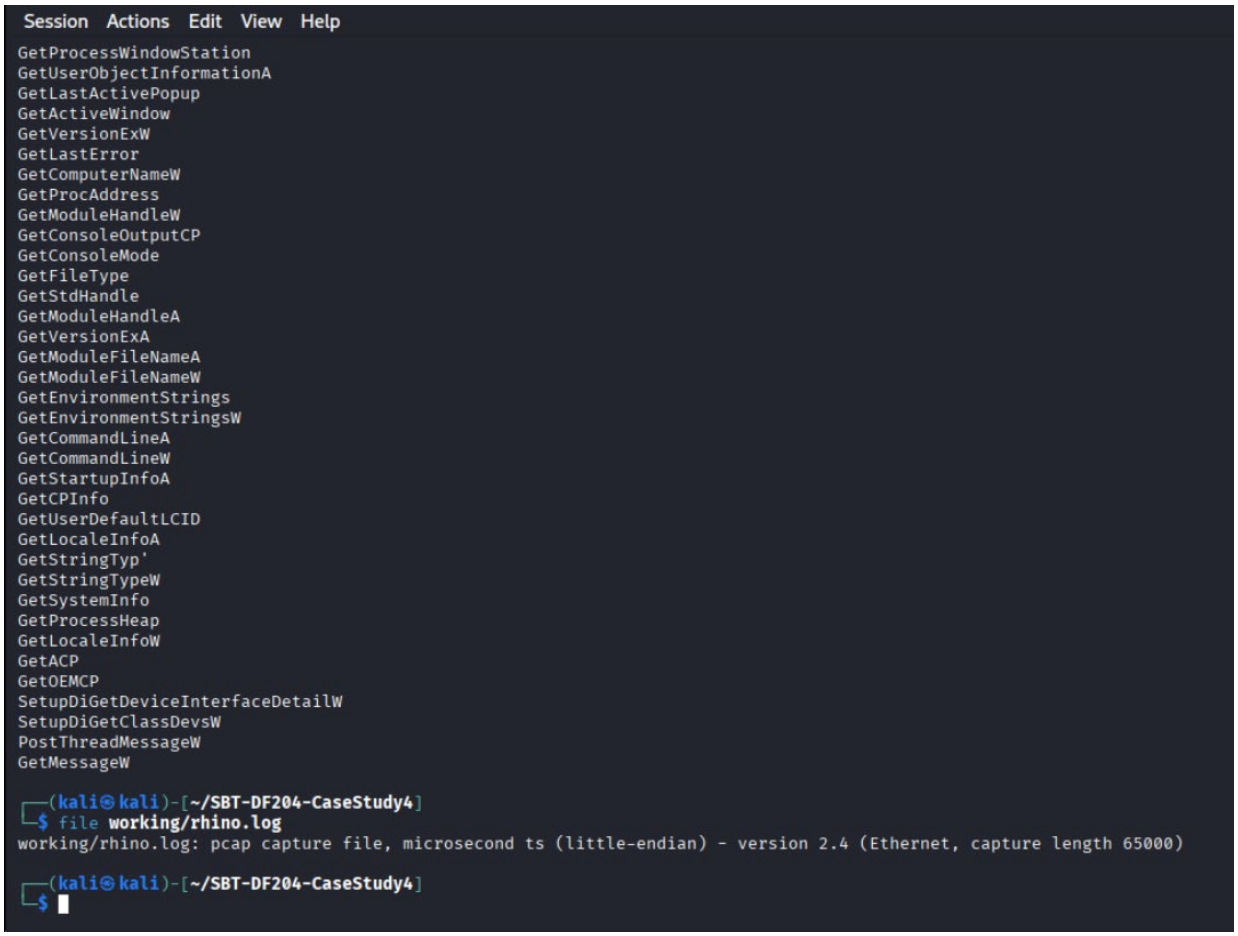


Figure 11: *PE32 Binary String & Import Dissection (rhino.exe)* — Extracting Win32 API function calls from rhino.exe and inspecting rhino.log pcap file structure.

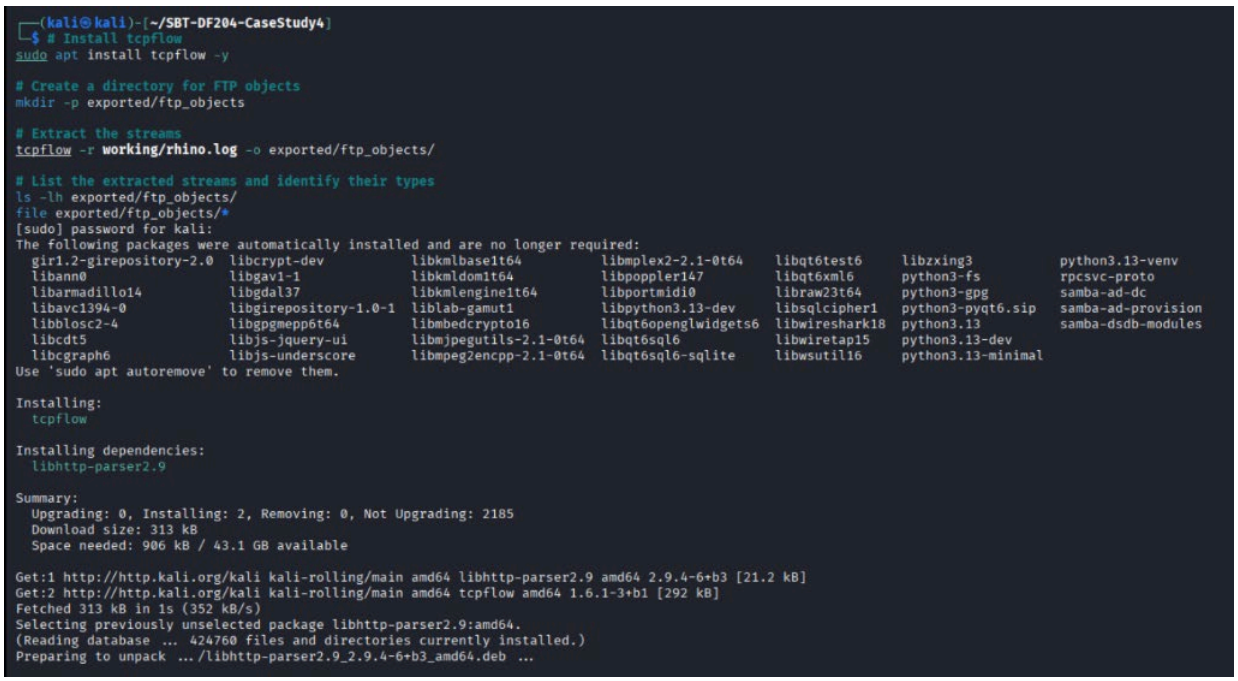


Figure 12: *FTP TCP Stream Carving via tcpflow* — Installing tcpflow and reconstructing TCP streams from rhino.log into exported/ftp_objects/.

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # Create a directory for HTTP objects
mkdir -p exported/http_objects

# Extract objects from rhino2.log
tshark -r working/rhino2.log --export-objects http,exported/http_objects/ -q

# Extract objects from rhino3.log
tshark -r working/rhino3.log --export-objects http,exported/http_objects/ -q

# List the extracted objects
ls -lh exported/http_objects/
file exported/http_objects/*
Warning: program compiled against libxml 215 using older 214
Warning: program compiled against libxml 215 using older 214
total 496K
-rw-r--r-- 1 kali kali 2.9K Oct 25 04:27  %2f
-rw-r--r-- 1 kali kali 2.9K Oct 25 04:27 '%2f(1)'
-rw-r--r-- 1 kali kali 2.9K Oct 25 04:27 '%2f(2)'
-rw-r--r-- 1 kali kali 2.9K Oct 25 04:27 '%2f(3)'
-rw-r--r-- 1 kali kali  216 Oct 25 04:27  back.gif
-rw-r--r-- 1 kali kali  148 Oct 25 04:27  blank.gif
-rw-r--r-- 1 kali kali  304 Oct 25 04:27 '~gnome'
-rw-r--r-- 1 kali kali  772 Oct 25 04:27 '~gnome(1)'
-rw-r--r-- 1 kali kali  309 Oct 25 04:27  image2.gif
-rw-r--r-- 1 kali kali 2.3K Oct 25 04:27  index.html
-rw-r--r-- 1 kali kali 8.4K Oct 25 04:27  logo.gif
-rw-r--r-- 1 kali kali 4.6K Oct 25 04:27  logo_sm.gif
-rw-r--r-- 1 kali kali  376 Oct 25 04:27  nav_current.gif
-rw-r--r-- 1 kali kali 1.1K Oct 25 04:27  nav_first.gif
-rw-r--r-- 1 kali kali  373 Oct 25 04:27  nav_page.gif
-rw-r--r-- 1 kali kali  290 Oct 25 04:27  object50
-rw-r--r-- 1 kali kali   89 Oct 25 04:27  object68
-rw-r--r-- 1 kali kali 1.4K Oct 25 04:27  object83
-rw-r--r-- 1 kali kali    5 Oct 25 04:27  object95
-rw-r--r-- 1 kali kali 1008 Oct 25 04:27  object97
-rw-r--r-- 1 kali kali 150K Oct 25 04:27  rhino4.jpg
-rw-r--r-- 1 kali kali  84K Oct 25 04:27  rhino5.gif
-rw-r--r-- 1 kali kali 143K Oct 25 04:27  rhino.exe
-rw-r--r-- 1 kali kali  15K Oct 25 04:27 'search%3fhl=en&ie=UTF-8&oe=UTF-8&q=rhino.exe'
-rw-r--r-- 1 kali kali  306 Oct 25 04:27 '~venkata'
-rw-r--r-- 1 kali kali 1.4K Oct 25 04:27 '~venkata(1)'
```

Figure 13: *HTTP Object Carving via TShark* — Exporting HTTP objects from rhino2.log and rhino3.log, recovering rhino4.jpg, rhino5.gif, and rhino.exe.

```
exported/http_objects/nav_page.gif:                                GIF image data, version 89a, 16 x 26
exported/http_objects/object50:                                    data
exported/http_objects/object68:                                    ASCII text, with CRLF line terminators
exported/http_objects/object83:                                    data
exported/http_objects/object95:                                    ASCII text, with CRLF line terminators
exported/http_objects/object97:                                    HTML document, ASCII text, with very long lines (407), with CRLF, LF line terminators
exported/http_objects/rhino4.jpg:                                  JPEG image data, JFIF standard 1.00, resolution (DPI), density 300×300, segment length
16, comment: "U-Lead Systems, Inc.", baseline, precision 8, 1531×2286, components 3
exported/http_objects/rhino5.gif:                                  GIF image data, version 89a, 400 x 275
exported/http_objects/rhino.exe:                                   PE32 executable for MS Windows 4.00 (console), Intel i386, 3 sections
exported/http_objects/search%3fhl=en&ie=UTF-8&oe=UTF-8&q=rhino.exe: HTML document, Unicode text, UTF-8 text, with very long lines (1930), with CRLF, LF lin
e terminators
exported/http_objects/~venkata:                                    HTML document, ASCII text
exported/http_objects/~venkata(1):                                 HTML document, ASCII text

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # Navigate to where your zip file is (rename the stream file to contraband.zip if needed)
# Example: mv exported/ftp_objects/192.168...  exported/ftp_objects/contraband.zip

# Check the zip file
file exported/ftp_objects/contraband.zip

# Attempt to extract using the discovered password
unzip -P gnome123 exported/ftp_objects/contraband.zip -d exported/extracted_contraband/
exported/ftp_objects/contraband.zip: cannot open `exported/ftp_objects/contraband.zip' (No such file or directory)
unzip:  cannot find or open exported/ftp_objects/contraband.zip, exported/ftp_objects/contraband.zip.zip or exported/ftp_objects/contraband.zip.ZIP.

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # Navigate to where your zip file is (rename the stream file to contraband.zip if needed)
# Example: mv exported/ftp_objects/192.168...  exported/ftp_objects/contraband.zip

# Check the zip file
file exported/ftp_objects/contraband.zip

# Attempt to extract using the discovered password
unzip -P gnome123 exported/ftp_objects/contraband.zip -d exported/extracted_contraband/
exported/ftp_objects/contraband.zip: Zip archive data, made by v2.0, extract using at least v2.0, last modified Apr 26 2004 17:00:12, uncompressed size 230
665, method=deflate
Archive:  exported/ftp_objects/contraband.zip
   skipping: rhino2.jpg              incorrect password

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$
```

Figure 14: *PE Executable File Verification & Failed Unzip Attempt* — Confirming rhino.exe is a PE32 executable, and demonstrating unzip failure with password 'gnome123'.

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy4]
└─$ # Search for keywords in the HTTP logs
strings working/rhino2.log working/rhino3.log | grep -iE "pass|zip|contraband|secret|rhino" | tee reports/http_password_search.txt
<IMG SRC="/icons/image2.gif" ALT="[IMG]"> <A HREF="rhino4.jpg">rhino4.jpg</A>             28-Apr-2004 16:09   150k
<IMG SRC="/icons/image2.gif" ALT="[IMG]"> <A HREF="rhino5.gif">rhino5.gif</A>             28-Apr-2004 16:09    83k
GET /~gnome/rhino4.jpg HTTP/1.1
GET /~gnome/rhino5.gif HTTP/1.1
<p> <a href="linkedlist.zip"> Linked List Implementation for HW 3</a></p>
GET /search?hl=en&ie=UTF-8&oe=UTF-8&q=rhino.exe HTTP/1.1
<html><head><meta HTTP-EQUIV="content-type" CONTENT="text/html; charset=UTF-8"><title>Google Search: rhino.exe </title><style><!—
</script><table border=0 cellpadding=4 cellspacing=0><tr><td class=q><font size=-1><font color=#000000><b>Web</b></font>    <a id=t1a c
lass=q href="http://images.google.com/images?q=rhino.exe&hl=en&lr=&ie=UTF-8&oe=UTF-8&sa=N&tab=wi" onClick="return qs(this);">Images</a>   &n
bsp;<a id=t2a class=q href="http://groups.google.com/groups?q=rhino.exe&hl=en&lr=&ie=UTF-8&oe=UTF-8&sa=N&tab=wg" onClick="return qs(this);">Groups</a> 
;   <a id=t4a class=q href="http://news.google.com/news?q=rhino.exe&hl=en&lr=&ie=UTF-8&oe=UT
F-8&sa=N&tab=wn" onClick="return qs(this);">News</a>     <a id=t5a class=q href="http://froogle.google.com/froogle?q=rhino.exe&hl=en&lr=
&ie=UTF-8&oe=UTF-8&sa=N&tab=wf" onClick="return qs(this);">Froogle</a><sup><a href="http://froogle.google.com/froogle?q=rhino.exe&hl=en&lr=&ie=UTF-8&oe=UTF
-8&sa=N&tab=wf" style="text-decoration:none;" onclick="return qs(this);"><font color=red>New!</font></a></sup>    </font><font size=-1>
<b><a href="/options/" class=q>more &raquo;</a></b></font></td></tr></table></td></tr><tr><td nowrap><form name=gs method=GET action=/search><input ty
pe=hidden name=hl value="en"><input type=hidden name=lr value=""><input type=hidden name=ie value="UTF-8"><input type=hidden name=oe value="UTF-8"><input t
ype=text name=q size=41 maxlength=2048 value="rhino.exe"><font size=-1> <input type=submit name="btnG" value="Search"><span id=hf></span></font></td><td no
wrap><font size=-2>  <a href=/advanced_search?q=rhino.exe&hl=en&lr=&ie=UTF-8&oe=UTF-8>Advanced Search</a><br>  <a href=/preferences?q=r
hino.exe&hl=en&lr=&ie=UTF-8&oe=UTF-8>Preferences</a>    </font></td></tr></table><table cellpadding=0 cellspacing=0 border=0><tr><td><f
ont size=-1></font></td></tr><tr><td height=7><img width=1 height=1 alt=""></td></tr></table></td></tr></form></table>
<table width=100% border=0 cellpadding=0 cellspacing=0><tr><td bgcolor=#3366cc><img width=1 height=1 alt=""></td></tr></table><table width=100% border=0 ce
llpadding=0 cellspacing=0 bgcolor=#e5ecf9><tr><td bgcolor=#e5ecf9 nowrap><font size=+1> <b>Web</b></font> </td><td bgcolor=#e5ecf9 align=right no
wrap><font size=-1 color=#000000>Results <b>1</b> - <b>10</b> of about <b>49</b> for <b><a href=/url?sa=X&oi=dict&q=http://dictionary.reference.com/search%
3Fq%3Drhino%26r%3D67 title="Look up rhino on dictionary.com"><b>rhino</b></a><b>.</b><a href=/url?sa=X&oi=dict&q=http://dictionary.reference.com/search%3Fq
%3Dexe%26r%3D67 title="Look up exe on dictionary.com"><b>exe</b></a></b>.  (<b>0.25</b> seconds) </font></td></tr></table><div><p class=g><a href=/url
?sa=U&start=1&q=http://www.pestpatrol.com/pestinfo%255Cs%255Csitescan.asp&e=747 >SiteScan</a><br><font size=-1> <b>...</b> Stop Running Processes: Kill the
se running processes with Task Manager:<br>
<b>rhino</b>.<b>exe</b> sitescan.exe. <b>...</b> <b>rhino</b>.<b>exe</b> sitescan.exe. Research. File Analyses: <b>...</b>
<br><font color=#008000>www.pestpatrol.com/pestinfo%5Cs%5Csitescan.asp -  30k - </font><a class=fl href=http://64.233.167.104/search?q=cache:FgIKA6r6OYgJ:w
ww.pestpatrol.com/pestinfo%255Cs%255Csitescan.asp+rhino.exe&hl=en&start=1>Cached</a> - <a class=fl href=/search?hl=en&lr=&ie=UTF-8&oe=UTF-8&q=related:www.p
estpatrol.com/pe
stinfo%255Cs%255Csitescan.asp>Similar pages</a></font> <p class=g><span class=w><font size=-2><b>[PDF]</b></font></span> <a href=/url?sa=U&start=2&q=h
ttp://www.bvsystems.com/Tech/Manuals/Rhino1.0.pdf&e=747 >BVS RHINO PC INTERFACE SOFTWARE</a><br><font size=-1><font color=#6f6f6f>File Format:</font> PDF/A
dobe Acrobat - <a href=http://64.233.167.104/search?q=cache:JakaEE7WfVEJ:www.bvsystems.com/Tech/Manuals/Rhino1.0.pdf+rhino.exe&hl=en&start=2>View as HTML</
a><br>Page 2. Page 3. BVS RHINO PC INTERFACE SOFTWARE INSTALLATION Copy the file &quot;<b>rhino</b>.<b>exe</b>&quot;<br>
<br><font color=#008000>www.bvsystems.com/Tech/Manuals/Rhino1.0.pdf -  </font><a class=fl href=/search?hl=en&lr=&ie=UTF-8&oe=UTF-8&q=related:www.bvsystems.
com/Tech/Manuals/Rhino1.0.pdf>Similar pages</a></font> <p class=g><a href=/url?sa=U&start=3&q=http://www.billybear4kids.com/jigsaw-puzzles/download-pg
2.html&e=747 >Jigsaw Puzzles from BillyBear4Kids.com</a><br><font size=-1> <b>...</b> Download 48 pc puzzle <b>rhino</b>.<b>exe</b> 9 pc puzzle <b>rhino</b
>.<b>exe</b> 130 pc puzzle h—<b>rhino</b>.<b>exe</b>.<br>
```

Figure 15: *HTTP Payload Keyword Search (Google Search: rhino.exe)* — Searching HTTP stream strings for pestpatrol malware references and Google search queries.

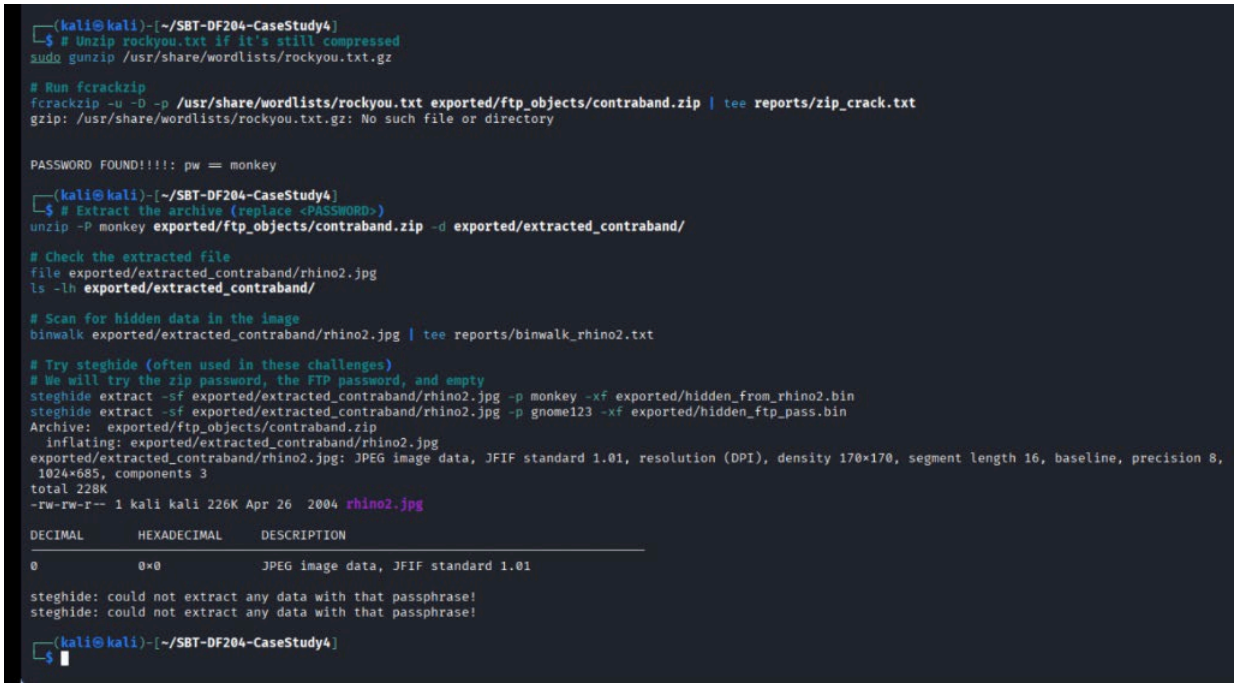


Figure 16: *Password Recovery via fcrackzip & Contraband Extraction* — fcrackzip cracking password 'monkey' from rockyou.txt, extracting rhino2.jpg (226K, 1024x685), and running steghide checks.

Student Declaration

> I confirm that I completed this investigation in an authorized environment, preserved the evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.
>
> Student Signature: Almustapha Yusuf Date: 26/10/2026